

Linear Algebra I · Course Update: Gap Analysis of New Assessments

Supplement to the existing 15-week course (Weeks 1–15). Sources analysed: 11 homework/assignment sets, 11 quizzes, 3 mock exams (25 files; 300 question items). **Tools: every item is by hand. No calculator and no Desmos** (all source exams state “no calculators”). No answer key included.

COVERAGE REPORT

Baseline (existing course). Concrete, geometric, $\mathbb{R}^n$ -centred: fields/complex numbers/axioms (W1), $\mathbb{F}^n$, $M_{n \times k}$, transpose, matrix products, $\mathcal{F}(\mathbb{F})$ (W2), systems/RREF (W3), sets/vectors/lines/planes (W4–5), span/LI (W6), dot product/projection (W7), subspaces/bases/dimension (W8), coordinates/change of basis (W9, W12), linear maps, composition, range/null space, rank-nullity, inverses (W10–12), determinant via volume (W13), eigen/diagonalization (W14), review (W15). **The new assessments are an abstract, proof-first version of the same syllabus** (general $\mathbb{F}$, $\mathbf{P}_n$, $M_{m \times n}$, $\mathcal{L}(V, W)$, matrices over $\mathbb{C}$, “prove/construct/justify”). Verified absences by keyword search: conjugate transpose, Cramer’s rule, multilinear/alternating determinant, Dimension Bound Theorem, isomorphism, $\mathcal{L}(V, W)$, direct sum, inner-product axioms, $\mathbb{F}^{\mathbb{N}}$ /shifts, P_n as a working space, subspace tests on matrix/polynomial/function constraint sets.

Counting convention. One *item* = one numbered problem (quizzes/mocks: each lettered part counted; homework: each numbered problem, sub-lists not expanded; definition-recall prompts excluded). *Unique* = first occurrence of a task in the set (order: HW→quiz of the same week → mocks); *Duplicate* = same task/technique already seen. Counts are by hand, ± 1 . *New concepts* = Category D; *new patterns* = Category C (Section “Integration Map” IDs).

File	Total	Unique	Dup.	New pat.	New conc.	Main new content
Assignment 1 (fields, VS axioms)	9	9	0	2	1	axiom-derived properties; $\mathbb{R}^+$ space
Quiz 1	6	0	6	0	0	clone of A1
Assignment 2 (subspaces, span)	9	9	0	3	2	A^* ; subspace tests in M , $\mathcal{F}(\mathbb{C})$
Quiz 2	6	0	6	0	0	clone of A2/A1
Assignment 3a (REF/RREF, solutions)	7	7	0	2	1	row-op equivalence (challenge)
Assignment 3b (LI/span tests)	8	7	1	4	0	over $\mathbb{C}$, M_2 , $\mathbf{P}_2$; pivot subsets
Quiz 3	8	2	6	2	0	$W_1 \cap W_2$; double inclusion
Quiz 4	12	1	11	1	0	RREF recognition
Assignment 4 (basis, dimension)	10	10	0	3	2	Dim. Bound, finite vs infinite dim.
Quiz 5	7	1	6	1	0	extend to basis
Assignment 6 (linear maps)	16	16	0	3	2	inner-product axioms, $W^\perp$, $W \oplus W^\perp$
Quiz 6	9	1	8	1	0	$\mathcal{L}(V, W)$ closure
Assignment 7 (ker/image)	8	8	0	4	1	ker/image via coordinates; $T(V')$, $T^{-1}(W')$
Quiz 7	11	2	9	2	0	construct trivial-kernel map
Assignment 8 (inj./surj., $\mathcal{L}$)	11	11	0	5	1	$\dim \mathcal{L}(V, W)$; shifts on $\mathbb{F}^{\mathbb{N}}$
Quiz 8	8	2	6	1	1	field-dependence (char. 2)
Assignment 9 (inverses)	11	11	0	4	1	isomorphism; $AB=I \Rightarrow BA=I$
Quiz 9	9	1	8	1	0	$T(R+S) = TR + TS$
Assignment 10 (determinants)	17	17	0	2	2	axiomatic det; Cramer
Quiz 10	8	0	8	0	0	clone of A10
Assignment 11 (eigen, diag.)	18	18	0	3	1	$p(A) = Pp(D)P^{-1}$
Quiz 11 (+4 survey parts)	8	0	8	0	0	clone of A11
Midterm 1 mock	26	7	19	5	0	“construct” prompts; identity-uniqueness
Midterm 2 mock (Part C placeholder only)	27	5	22	5	0	$(AB)^*$; associativity; LT from relations
Final mock (Part C unreadable)	31	12	19	8	1	conjugate eigenpairs; basis criterion
Total	300	157	143	62	16	

Finding 1. Quizzes are $\approx 90\%$ clones of their homework (82 of 92 quiz items are duplicates). **Homework is the source of value; quizzes add only 10 items.** **Finding 2.** After merging, the 157 first-occurrence items collapse to **83 integration entries** (52 Core, 8 Optional, 4 Enrichment, 9 Exam Challenge, 10 Mock Exam Material). **Finding 3.** No file assesses least squares, Gram–Schmidt, orthogonal diagonalization, LU, or complex inner products; none is added. **Gaps in sources:** Mock Midterm 2 Part C (“group”) is a placeholder; Mock Final Part C has no question text; HW2 T_5, T_6 are garbled in the source and omitted. The first Part-B question in quizzes is printed “D1” (typo for B1).

Category A (already covered, no action): RREF reduction and back-substitution (W3), $2 \times 2/3 \times 3$ determinant

formulas (W13), matrix of a map on $\mathbb{R}^n$ (W10), inverse by row reduction (W12), characteristic polynomial of triangular/small matrices (W14), dot product/projection formula (W7), span/LI over $\mathbb{R}^n$ (W6), rank–nullity statement (W11). **Category B (better example, enrichment):** complex-entry RREF/inverse/eigen problems (Q4 D1, MT2 B7, Final B6, Q11 B2); $p(D)$ for diagonal D ; triangle/tetrahedron volume by determinant.

NEW MATERIAL SUMMARY

Category D — new concepts (16)

Topic	New Element	Why It Matters	Recommended Location
D1 Conjugate transpose	$A^* = \overline{A}^T$; $(X+Y)^*$, $(cX)^* = \bar{c}X^*$, $(AB)^* = B^*A^*$; expand/compute/prove	Tested in Q2, MT1, MT2, Final setup; sign of i is the trap	W2 after transpose
D2 Abstract carriers	$\mathbf{P}_n(\mathbb{F})$, $M_{m \times n}(\mathbb{F})$ as spaces for subspace/span/LI/basis/dim/maps	Most exam items live here, not in $\mathbb{R}^n$	W2 (intro), then W6, W8, W10
D3 Dim. Bound & Extension Lemma	Named theorems; “ $ S > \dim V \Rightarrow \text{LD}$ ”, “ $ S < \dim V \Rightarrow \text{not spanning}$ ”, “ $ S = \dim V \text{ LI} \Rightarrow \text{basis}$ ”	Replaces long computations by 2-line counting proofs (HW4, Q5, MT1 B3, Final B9)	W8
D4 Finite vs infinite dim.	$\mathbf{P}(\mathbb{R})$ not finite-dim.; $\mathbb{F}^{\mathbb{N}}$; left/right shift; subspaces of infinite-dim. spaces	Shows injective and surjective differ for operators in infinite dimension	W8 (dim.), W12 (shifts)
D5 $\mathcal{L}(V, W)$	It is a vector space; $\dim \mathcal{L}(V, W) = \dim V \cdot \dim W$; maps out of $\mathcal{L}$	Q6, HW8, Final A10	W11
D6 Isomorphism	Bijjective linear map; $V \cong W \iff \dim V = \dim W$; $W \cong \mathbb{F}^n$	MT2 B8(b), HW9; bridges abstract $\leftrightarrow \mathbb{F}^n$	W12
D7 Axiomatic determinant	Unique alternating multilinear map, $\det I = 1$; compute/prove from axioms (no $ad - bc$)	Q10, A9/B5 Final force it; course only has volume definition	W13 (new subsection)
D8 Cramer’s rule	$x_i = \det A_i / \det A$, applicability iff $\det A \neq 0$	Q10 B4; 7 HW systems incl. singular traps	W13
D9 $p(A)$, $p(T)$	$A^r = MB^rM^{-1}$, $p(A) = Mp(B)M^{-1}$, $p(D) = \text{diag } p(d_i)$	Q11, Final A5, B6(e,f)	W14
D10 Inner-product axioms, $W^\perp$, $\oplus$	Axioms (additivity, homogeneity, symmetry, positive-definite); $W^\perp$ subspace; $W \oplus W^\perp = \mathbb{R}^n$	HW6 only (challenge); proofs not in W7	W7 (optional)
D11 Properties from axioms	Uniqueness of 0/inverse, cancellation, $0v = c0 = 0$, $(-1)x = -x$; field analogues	Part A/B recall in Q2, MT1 A9, Final B4	W1 (extend “Proof Techniques”)
D12 Operator catalogue	identity, zero, integration $\mathbf{P}_n \rightarrow \mathbf{P}_{n+1}$, derivative, $A \mapsto A^T$, $A \mapsto A + A^T$; matrices/eigenvalues	Rekurs in HW6, 7, 9, 11, Q6–Q9	W10–W12
D13 Field dependence	Same formula over $\mathbb{C}$ vs $\mathbb{F}$; $(x+y, y+z, x+z)$ bijective iff $2 \neq 0$	Q8 B2(b)	W12
D14 Conjugation and eigen	$\bar{z}\bar{w} = \overline{zw}$; $\bar{A}\bar{v} = \overline{Av}$; real A : $(\lambda, v) \Rightarrow (\bar{\lambda}, \bar{v})$	Final B2; explains complex pairs	W14
D15 Row-op equivalence	Each elementary row operation preserves the solution set	Challenge HW3a (not on quizzes)	W3 (optional)
D16 Image/preimage of subspaces	$T(V')$, $T^{-1}(W')$ are subspaces; $T^{-1}(W') = \ker T$ choice	HW7; feeds $\ker T = \ker T^2$	W11

Category C — new question patterns (merged into lessons; detail in Bank)

- **“Construct an object with property” (Part A).** Subspace of $M_2(\mathbb{C})$ of dimension 2; spanning-but-dependent subset of $\mathbf{P}_3$; map $\mathbf{P}_2 \rightarrow \mathbb{R}^3$ not iso.; A, B singular with $A + B$ invertible; $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with one eigenvalue; inconsistent REF. W8, W10, W12, W14
- **“Set up only” with if-and-only-if justification.** LI / span-membership / spanning-set systems over $\mathbb{C}$, M_2 , $\mathbf{P}_2$. W6
- **Subspace test with constraint sets + counterexample** ($f(0) = 1$, $p(1) = 1$, $\det A = 0$, $A_{11} = 1$, $x + y - z = 2$). W8
- **Kernel/image of abstract maps via coordinate matrix** (nullspace $\rightarrow \ker$, pivot columns $\rightarrow \text{image}$), with rank–nullity check. W11
- **Yes/No dimension reasoning** (surjective? injective? from dim alone). W11
- **Linear map from relations** ($T(1, 2, 3)$, $T(1, 1, 1)$ given; find $T(0, 1, 2)$). W10
- **Basis extension and basis-criterion proofs.** W8
- **Compute-and-verify coordinate identity** $[T(v)]_\beta = [T]_\alpha^\beta [v]_\alpha$ with non-standard β . W9
- **Change-of-basis identities** $[I]_\gamma^\alpha = [I]_\beta^\alpha [I]_\gamma^\beta$, $[I][I] = I$, similarity of representations. W12
- **Entrywise matrix-algebra proofs** (sum notation). W2

QUESTION INTEGRATION MAP

IDs are used in the Bank. Source: A# = assignment/homework, Q# = quiz, MT1/MT2 = mock midterms, F = mock final. Difficulty E/M/H. Classification: Core Addition, Optional Practice, Exam Challenge, Mock Exam Material (Enrichment = Category B, counted under Optional).

ID	Source	Topic	Diff	Reason for Inclusion	Class
RR1	Q4 A1	RREF recognition	E	Pivot-column zero rule is tested directly	Core
RR2	MT1 A6, A3a	Construct inconsistent REF; classify systems	E	Recurring “construct” prompt	Core
RR3	A3a (ch.)	Row operations preserve solution set	H	Justifies method; never quizzed	Optional
RR4	A3b, Q4 A3, Q5 A2	Column/row bases from RREF, incl. complex	M	Col basis = original pivot columns	Core
RR5	A3b	Basic-variable theorem (pivot vectors span, LI)	H	Proof behind RR4/BD6	Core
LS1	A3a	One system family, five tasks (REF, RREF, basic/free, solution space, affine set)	M	Instructor’s drill format	Core
LS2	Q3 B2, Q4 D1, B2	Set-up-only systems with iff justification	M	Marks for the justification	Core
LS3	Q3 B3, Q4 B3	Solution-set subspaces; spanning set; double inclusion	M	Constraint $\rightarrow$ span	Core
LS4	MT1 C1	Homogeneous solution set is a subspace	H	Extension question	Exam Challenge
LS5	A8, Q8 D1	$Ax = b$ over $\mathbb{C}$ (triangular)	M	Complex back-substitution	Core
MO1	A2, Q2, MT1 A4, MT2 A6	Conjugate-transpose expand/compute	M	Sign of $\bar{i}$ trap	Core
MO2	A2, Q2 B3, MT2 B2	Prove/verify $(X+Y)^*$, $(cX)^*$, $(AB)^*$	M	Entrywise index proof	Core
MO3	Q9 A1, MT2 A2, F B10	Complex matrix products	M	$i^2 = -1$ slips	Core
MO4	Q6 D1	4×4 complex matrix–vector product	M	Arithmetic only	Enrichment
MO5	A7, Q7 D1, Q9 D1	Entrywise distributivity proofs	E	Sum-notation template	Core
MO6	MT2 A10	Singular A, B with $A + B$ invertible	E	Invertibility not additive	Mock Exam
MO7	A9, Q9 A2	Invertibility of $2 \times 2/3 \times 3$ over $\mathbb{C}$	M	Determinant or rank check	Core
MO8	A1	Function algebra and composition notation	E	Fixes fg vs $f \circ g$	Enrichment
DT1	A10, F A9, B5	Determinant from axioms (no $ad - bc$)	H	Explicitly required on exams	Core
DT2	A10	Induction proofs: $\det A^T$, triangular, $\det kA = k^n \det A$, row ops	H	Proof bank for DT3	Optional
DT3	Q10 B3	Scaling a row scales det	M	Use multilinearity only	Core
DT4	A10, Q10 B2	Cofactor 3×3 , 4×4 over $\mathbb{C}$	M	Sign pattern	Core
DT5	A10	Area/volume by det; triangle $\frac{1}{2}$, tetrahedron $\frac{1}{6}$	M	Application	Optional
DT6	A10, Q10 B4	Cramer’s rule with singular traps	M	Applicability check is marked	Core
DT7	Q10 A3	State characterization of det	E	Part-A recall	Core
VS1	A1, Q2, MT1 A9, F B4a	Properties from axioms	M	Repeated on every exam type	Core
VS2	MT1 B2	Identity unique from partial axioms	M	Needs commutativity step	Mock Exam
VS3	A1, Q1 B3, MT1 B1	Axiom check in $\mathbb{F}^n$, M , $\mathcal{F}(\mathbb{C})$	E	Pointwise/entrywise template	Core
VS4	A1 (ch.)	$\mathbb{R}^+$ with $\boxplus = \cdot$	H	Non-standard operations	Exam Challenge
VS5	Q1, MT1 A1, A1	$\mathbb{C}$ closure, identity, arithmetic	E	Week-1 quiz pattern	Core
VS6	A2	Subspace lists: matrices, functions	M	Broad constraint variety	Core
VS7	A3a, Q2, MT1 B5, F B1	Constraint-set tests incl. $\det A = 0$	M	Zero-vector disproof	Core
VS8	Q3 D1	$W_1 \cap W_2$ subspace	M	Not in existing W8	Core
VS9	A2, MT1 B4	Span properties	M	Set-inclusion proofs	Core
VS11	A2	True/false/sometimes on LI/LD inclusion	M	Reasoning drill	Optional
VS12	MT1 A8, A10	Construct subspace dim 2; dependent spanning set	M	Part-A construction	Mock Exam
VS13	A4	Finite vs infinite dimension; $\mathbf{P}(\mathbb{R})$	H	Infinite-dim. logic	Exam Challenge
VS14	A9, MT2 B8	$W \cong \mathbb{F}^n$; symmetric matrices	E	Dimension as invariant	Core
LI1	A3b, Q3, Q4	LI tests: $\mathbb{C}^3$, $M_2(\mathbb{C})$, $\mathbf{P}_2$	M	Complex + abstract	Core
LI2	A3b, Q3, Q4, MT1	Span-membership tests	M	Coefficient matching	Core
LI3	A2	$\{u, v, w\}$ LI iff $\{u+v, u+w, v+w\}$ LI	H	Complex field subtlety	Exam Challenge
LI4	A2	Proper subset of LI set has smaller span	M	Uses minimality	Optional
LI5	MT1 B6	Full LI workflow + basis extraction	M	Mock pattern	Mock Exam

ID	Source	Topic	Diff	Reason for Inclusion	Class
LI6	F B8, A6, A8	$T(v_i)$ LI $\Rightarrow v_i$ LI; preservation	M	Converse direction is tested	Core
BD1	A4	Standard bases of $\mathbb{F}^n$, $M_{n \times k}$, $\mathbf{P}_n$; dimensions	M	Needed everywhere	Core
BD2	A4, Q5, MT1, F B1b	Basis/dim of constraint subspaces	M	Highest-frequency mark	Core
BD3	A4, Q5 B2, MT1 B3, A5	Counting arguments (Dim. Bound)	M	Two-line proofs	Core
BD4	Q5 B3	Extend LI set to basis	M	Must justify choice	Core
BD5	F B9	Basis criterion $ S = n$	H	Uses Extension Lemma	Exam Challenge
BD6	A3b	Extract LI subset with same span	M	Pivot vectors	Core
LT1	A6, Q6 B3	Linearity tests incl. $A+A^T$, $\text{tr}+1$, $A+I$	M	Zero-vector disproof	Core
LT2	A6, Q6 B4b, MT2 B6	Properties: $T(0)$, $T(-v)$, image subspace	E	Foundational proofs	Core
LT3	A6, Q6 B2, Q7, MT2	Matrix reps on abstract/non-standard bases	M	Coordinate verification	Core
LT4	MT2 A7	Matrix $\rightarrow$ explicit T	E	Part-A pattern	Mock Exam
LT5	MT2 A9	T from relations (non-basis)	M	Write as combination	Mock Exam
LT6	A6	Identity, zero, integration operators	E	Composition identities	Optional
LT8	A7, MT2 B3	Kernel/image via coordinate matrix; rank-nullity	H	Full 6-part mark block	Core
LT9	A7	$\text{im } T = \ker T$ possible iff $\dim V$ even	H	Dimension reasoning	Exam Challenge
LT10	A7, MT2 B5	$\ker T = \ker T^2 \iff \text{im } T = \text{im } T^2$	H	Rekurs in HW and mock	Exam Challenge
LT11	Q7 A1–2, MT2 A5, F A1	Construct maps with stated property	E	Part-A construction	Mock Exam
LT12	A8, Q8, MT2, F A10	Dimension yes/no; $\dim \mathcal{L}(V, W)$	E	Rapid Part-A marks	Core
LT13	A8, Q8	Inj./surj. properties; shifts	M	Infinite-dim. counterexamples	Core
LT14	Q6 B4a, Q9 B2, MT2 B4	$\mathcal{L}(V, W)$ closure; distributivity; associativity	M	Pointwise equality	Core
LT15	A9, Q9 B4	ST inj./surj. consequences; iso example	H	Needs counterexample	Exam Challenge
INV1	A9, Q9 B3, MT2 B7	Inverse via chosen bases; explicit T^{-1}	H	Complex 4×4	Core
INV2	A9	$AB = I \Rightarrow BA = I$; uniqueness; T^{-1} linear	M	Proof block	Core
INV3	Q8 B2b	Field dependence (char. 2)	H	Unusual trap	Optional
EV1	A11, Q11 B3, F A5, B6e	$p(A), p(T)$ via diagonalization	M	Reused on three assessments	Core
EV2	A11	Eigen/diagonalization drill (6+4 matrices)	M	Exam-format matrices	Core
EV3	A11	Eigen of derivative, transpose, shear	H	Abstract operators	Core
EV4	Q11, A11	Eigen over $\mathbb{C}$ (triangular)	M	Read diagonal	Core
EV5	F A1, A4, A6–8	Conceptual eigen (invertible, E_0)	E	Part-A recall	Mock Exam
EV6	F B2, B4b	Conjugate eigenpair; $E_{\lambda_1} \cap E_{\lambda_2} = 0$	M	Two short proofs	Core
EV7	F B6	Diagonalizable T , non-diagonalizable S over $\mathbb{C}$	H	Geometric multiplicity	Mock Exam
EV8	F B7	3×3 char. poly. with complex roots	M	Rational-root then quadratic	Mock Exam
EV9	A11	Orthogonal projection eigenvalues 0, 1	M	Geometric link	Enrichment
CB1	A11, Q11 D1, F B3	Change-of-basis identities	M	Composition proofs	Core
CB2	A11	Computations on $\mathbb{C}^2, \mathbb{C}^3, \mathbf{P}_2$	M	Non-standard bases	Core
OR1	A6	Inner-product axioms; multilinearity	E	Axiom template	Optional
OR2	A6	$W^\perp$ subspace; $W \oplus W^\perp = \mathbb{R}^n$	H	Direct sum new	Exam Challenge
OR3	A6, A9, Q6 A2	P_v , reflection, rotation as operators	M	Properties $P^2 = P$, $R^2 = I$	Core
OR4	A6	12 projection/rotation computations	E	Fluency	Enrichment
OT1	All Part A	Definition/theorem recall list (see High-Yield)	E	One mark each, every quiz	Core

SUPPLEMENTAL PROBLEM BANK

All by hand (no calculator, no Desmos). Questions only; no answers. “Show/justify” = full proof or working expected. $\mathbb{F}$ is an arbitrary field unless stated.

1. Row Reduction

- RR1** Which of $A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ are in RREF? For each other matrix, name the violated condition.
- RR2** (a) Write an augmented matrix in REF of an inconsistent system. (b) Write one with a unique solution and one with infinitely many. (c) For $\begin{bmatrix} 1 & 2 & -1 & 0 & 5 \\ 0 & 1 & 3 & 4 & -2 \\ 0 & 0 & 0 & 1 & 7 \end{bmatrix}$ state the basic and free variables.
- RR3** (*challenge*) Let S' arise from S by one elementary row operation ($R_i \leftrightarrow R_j$, cR_i , $R_i + \lambda R_j$). Prove x solves $S \iff x$ solves S' ; conclude S and S_{rref} have equal solution sets.
- RR4** Reduce to RREF and give LI spanning sets for $\text{col}(A)$ and $\text{row}(A)$: (a) $\begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ 1 & 1 & 0 \end{pmatrix}$, (b) $\begin{pmatrix} 2 & 1 & 0 & 3 \\ 1 & 0 & 1 & 2 \\ 3 & 1 & 1 & 5 \end{pmatrix}$,
(c) $\begin{pmatrix} 1 & i & 0 \\ 2 & 2i & 1 \\ 1 & i & 1 \end{pmatrix}$ over $\mathbb{C}$.
- RR5** Let the homogeneous system for $a_1v_1 + \dots + a_nv_n = 0$ reduce to REF with basic indices B , free indices F . Prove (a) $v_f \in \text{span}\{v_b : b \in B\}$ for $f \in F$; (b) $\text{span}\{v_b\} = \text{span}(S)$; (c) $\{v_b : b \in B\}$ is LI. Deduce $\text{col}(A) = \text{span}\{c_i : i \text{ pivot column of } \hat{A}\}$.

2. Linear Systems

- LS1** For each system, (i) Gaussian elimination to REF, (ii) Gauss–Jordan to RREF, (iii) basic/free variables, (iv) for the homogeneous version a spanning set of the solution space, (v) for the non-homogeneous version the solution set as $p + \text{span}(\dots)$:
(a) $x+y+z = 2$, $2x+2y+2z = 4$, $x-y+z = 0$; (b) $x+y+z = 3$, $2x-y+z = 1$, $-x+y+2z = 4$; (c) $x+y = 2$, $2x-y = 1$, $3x = 3$; (d) $x+y+z = 1$, $2x+2y+2z = 2$, $x+y+z = 3$.
- LS2** *Set-up only.* Let $v_1 = (1, i, 0, 1)$, $v_2 = (2, 1, i, 0)$, $v_3 = (3, 1+i, i, 1) \in \mathbb{C}^4$. (a) Using an iff chain, construct a system whose solutions decide if $\{v_1, v_2, v_3\}$ is LI. (b) Likewise decide if $w = (4, 2+i, i, 1) \in \text{span}\{v_1, v_2, v_3\}$.
- LS3** (a) $W = \{(a, b, c, d) \in \mathbb{R}^4 : a + b + c + d = 0\}$: build a system, solve by Gauss–Jordan, write $W = \text{span}(S)$ with S LI. (b) $W = \{(x, y, z) : 2x + y = z\}$: construct S and prove $\text{span}(S) = W$ by double inclusion.
- LS4** Let x, y solve $a_{r1}v_1 + \dots + a_{rn}v_n = 0$ ($r = 1..n$). (a) Show $cx + y$ solves a generic row. (b) Deduce the solution set is a subspace of $\mathbb{F}^n$.
- LS5** Solve $Ax = b$ in $\mathbb{C}^3$: (a) $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1-i & 1 \\ 0 & 0 & i \end{pmatrix}$, $b = (i, 1, 1+i)^T$; (b) $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & i \\ 0 & 0 & 0 \end{pmatrix}$, $b = (1+i, 2, 0)^T$; (c)
 $A = \begin{pmatrix} 1+i & 1 & 0 \\ 0 & 1 & i \\ 0 & 0 & 1 \end{pmatrix}$, $b = (2, 1+i, -i)^T$.

3. Matrix Operations (including inverses)

- MO1** For $A, B \in M_{n \times k}(\mathbb{C})$ expand with no brackets: (a) $(2A - 3B)^*$, (b) $((3-i)A + 2B)^*$, (c) $3iA(iB - 4C)^*$. (d) Compute $(2A - 3B)^*$ for $A = \begin{pmatrix} 1 & 2 \\ -i & 3 \end{pmatrix}$, $B = \begin{pmatrix} 0 & -1 \\ 2 & i \end{pmatrix}$.
- MO2** Prove for $X, Y \in M_{m \times n}(\mathbb{C})$, $c \in \mathbb{C}$: $(X+Y)^T$, $(X+Y)^*$, $(cX)^* = \bar{c}X^*$, and for $A \in M_{n \times l}$, $B \in M_{l \times k}$: $(AB)^* = B^*A^*$. Verify the first three for $A = \begin{pmatrix} 1+i & 2 \\ -3i & 4 \end{pmatrix}$, $B = \begin{pmatrix} 2 & -1+i \\ 5 & 3i \end{pmatrix}$, $c = 2-i$.
- MO3** Compute: (a) $\begin{pmatrix} 1+i & 2 \\ 0 & 1-i \end{pmatrix} \begin{pmatrix} i & 1 \\ 2 & -i \end{pmatrix}$; (b) $(AB)^*$ for $A = \begin{pmatrix} 3i & 4 \\ 2-i & 2 \end{pmatrix}$, $B = \begin{pmatrix} 1 & -i \\ 0 & 2i \end{pmatrix}$; (c) $\begin{pmatrix} 1+i & 2 & 0 \\ -i & 1 & 3 \\ 2 & i & 1-i \end{pmatrix} \begin{pmatrix} 2-i & 1 \\ 1+i & 0 \\ i & 1-i \end{pmatrix}$
- MO4** (*enrichment*) $\begin{pmatrix} 1+i & 2 & 0 & -i \\ 0 & 1-i & 3i & 2 \\ 2 & 0 & i & 1 \\ -i & 1 & 0 & 2+i \end{pmatrix} \begin{pmatrix} 1 \\ i \\ -1 \\ 2 \end{pmatrix}$.
- MO5** For $A, B \in M$ with compatible shapes prove entrywise (write $(Au)_{i1} = \sum_j a_{ij}u_{j1}$): $A(u+v) = Au + Av$; $(cA)u = c(Au) = A(cu)$; $A0 = 0$; $A(cu+dv) = cAu + dAv$; $I_n u = u$; $(A+B)u = Au + Bu$; $A(B+C) = AB + AC$.

- MO6** Construct $A, B \in M_2(\mathbb{R})$, both non-invertible, with $A + B$ invertible.
- MO7** Is it invertible? (a) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, (b) $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, (c) $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$, (d) $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{pmatrix}$, (e) $\begin{pmatrix} 1+i & 2 \\ 3 & 1-i \end{pmatrix}$.
- MO8** (*enrichment*) $f(x) = x^2 + 3x$, $g(x) = 2x + x^2$. Compute $f(2x+3)$, $(2f+3)(x)$, $(2f+2g)(x)$, $(fg)(x)$, $(f \circ g)(x)$. State when juxtaposition means product and when composition.
- INV1** For each T : choose ordered bases, compute the matrix, use Gauss–Jordan to decide invertibility, and if invertible give T^{-1} explicitly; else say why not. (a) $T(x, y) = (2x+y, x-y)$; (b) $p \mapsto p'$ on $\mathbf{P}_2(\mathbb{R})$; (c) $A \mapsto A + A^T$ on $M_2(\mathbb{R})$; (d) $T(x, y, z) = (x+y, y+z, x+z)$; (e) $a+bx+cx^2 \mapsto a+2bx+3cx^2$; (f) $p \mapsto \int_0^x p$, $\mathbf{P}_2 \rightarrow \mathbf{P}_3$; (g) $A \mapsto \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} A$; (h) $(x, y, z) \mapsto (x, y, 0)$; (i) $a+bx+cx^2 \mapsto a + (b+a)x + cx^2$; (j) $T : M_2(\mathbb{C}) \rightarrow \mathbf{P}_3(\mathbb{C})$, $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \mapsto (a+bi) + (b+ci)x + (c+di)x^2 + dx^3$ (find $([T]_\alpha^\beta)^{-1}$ and T^{-1}).
- INV2** (a) If $AB = I_n$ in $M_{n \times n}(\mathbb{F})$, explain via linear maps why $BA = I_n$. (b) Prove an inverse of a linear map is unique; same for matrices. (c) Prove T^{-1} is linear when T is an invertible linear map.
- INV3** $T(x, y, z) = (x+y, y+z, x+z)$ on $\mathbb{F}^3$. Without computing ker or image, decide injectivity and surjectivity; how does the answer depend on $\mathbb{F}$? (Hint: $\det = 2$.)

4. Determinants

- DT1** Using only multilinearity, alternation and $\det(e_1, \dots, e_n) = 1$ (no $ad - bc$, no cofactors) compute $\det$ of (a) $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, (b) $\begin{pmatrix} 2 & 3 \\ 2 & -4 \end{pmatrix}$, (c) $\begin{pmatrix} 1 & i & 0 \\ 2 & 3 & 0 \\ 0 & 0 & 2-i \end{pmatrix}$, (d) $\begin{pmatrix} 1+i & 1 & 0 \\ 0 & 1 & i \\ 0 & 0 & 2 \end{pmatrix}$, (e) $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{pmatrix}$.
- DT2** Prove (induction where indicated): $\det A^T = \det A$; $\det \bar{A} = \overline{\det A}$ (challenge) hence $\det A^* = \overline{\det A}$; triangular $\Rightarrow \det A = \prod a_{ii}$; $\det kA = k^n \det A$; equal rows $\Rightarrow 0$; swapping rows negates; adding a multiple of a row leaves $\det$ unchanged.
- DT3** Using only the multilinear and alternating properties prove: if B is A with row i multiplied by k , then $\det B = k \det A$.
- DT4** Cofactor expansion: (a) $\begin{pmatrix} 2 & 1 & 0 \\ -1 & 3 & 4 \\ 0 & 2 & 5 \end{pmatrix}$, (b) $\begin{pmatrix} 1 & 0 & 2 \\ 3 & 1 & 4 \\ 2 & 0 & 1 \end{pmatrix}$, (c) $\begin{pmatrix} 1+i & 1 & 2 \\ 2 & 1+i & 1 \\ 1 & 2 & 1+i \end{pmatrix}$, (d) $\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 4 \end{pmatrix}$.
- DT5** (a) Area of the triangle $(1, 2), (4, 1), (6, 5)$; of $(-1, 3), (2, 0), (5, 4)$. (b) Area of the parallelogram spanned by $(2, -1), (4, 3)$. (c) Volume of the parallelepiped spanned by $(1, 0, 2), (2, 1, 1), (3, -1, 4)$. (d) Volume of the tetrahedron $(0, 0, 0), (1, 2, 0), (3, 1, 1), (2, 0, 4)$. (e) Explain the factors $\frac{1}{2}$ and $\frac{1}{6}$.
- DT6** Decide whether Cramer's rule applies (show $\det \neq 0$); if so solve: (a) $x+2y = 5$, $3x+4y = 11$; (b) $2x-y = 1$, $4x-2y = 2$; (c) $x+y+z = 6$, $2x+3y+z = 10$, $x+4y+5z = 20$; (d) $x+2y+z = 4$, $2x+4y+2z = 8$, $x+y+z = 3$; (e) $3x-y+2z = 7$, $2x+y-z = 1$, $x+4y+z = 9$; (f) $x+y = 2$, $2x+2y = 4$, $3x+3y = 6$; (g) $2x+y-z = 3$, $x-y+2z = 0$, $3x+2y+z = 5$; (h) $x+y = 3$, $2y+z = 4$, $x+z = 5$.
- DT7** State the characterization of $\det : M_{n \times n}(\mathbb{F}) \rightarrow \mathbb{F}$ by its multilinear and alternating properties.

5. Vector Spaces

- VS1** Prove for a field: additive identity unique; additive inverse unique; $0 \cdot c = 0$; $(-1)c = -c$. Prove for a vector space V : zero vector unique; inverse unique; cancellation ($x+y = x+z \Rightarrow y = z$); $0v = c0 = 0$; $(-1)x = -x$. List two such properties that are not axioms.
- VS2** $\boxplus : S \times S \rightarrow S$ is commutative and has 0 with $0 \boxplus s = s$ for all s . Prove 0 is the unique such element.
- VS3** Prove: (a) $c(x+y) = cx + cy$ in $\mathbb{F}^n$; (b) $(c+d)f = cf + df$ in $\mathcal{F}(\mathbb{C})$; (c) $M_{n \times k}(\mathbb{C})$ and $\mathcal{F}(\mathbb{C})$ are vector spaces; (d) $\mathcal{F}_{\text{even}}(\mathbb{R})$ is a vector space. Structure: given / to prove / technique.
- VS4** (*challenge*) On $\mathbb{R}^+ = \{x > 0\}$ set $x \boxtimes y = xy$, $c \boxdot x = x^c$. Show $(\mathbb{R}^+, \boxdot, \boxtimes)$ is an $\mathbb{R}$ -vector space; identify its zero and additive inverses.
- VS5** (a) Prove $\mathbb{C}$ is closed under $+$ and $\times$ and has an additive identity. (b) Compute $(2-3i)(1+2i)$; $1/i$; $(1+i)^{-1}$; $\frac{3+2i}{1-i}$; $-(2-5i)$; $(2+i)^{-1}$.
- VS6** Subspace of $M_{m \times n}(\mathbb{F})$ (proof or counterexample): $A_{11} = 0$; $A_{11} = 1$; $A_{12} = 2A_{21}$; symmetric; skew-symmetric; diagonal. Subspace of $\mathcal{F}(\mathbb{C})$: $f(0) = 0$; $f(0) = 1$; even; odd; real-valued.
- VS7** Subspace? If yes prove it; if not give a counterexample: (a) $\{(x, y, z) : x+y-z = 0\}$; (b) $\{A \in M_2(\mathbb{C}) : A = A^*\}$; (c) $\{p \in \mathbf{P}_2(\mathbb{R}) : p(1) = 1\}$; (d) $\{A \in M_2(\mathbb{C}) : \text{tr} A = 0\}$; (e) $\{p \in \mathbf{P}_3(\mathbb{R}) : p(0) + p(1) = 0\}$; (f) $\{x+y-z = 2\}$; (g) $\{f \in \mathbf{P}_2(\mathbb{C}) : f(1) = f(2), f(3) = 1\}$; (h) $\{A \in M_2(\mathbb{R}) : \det A = 0\}$; (i) $\{f \in \mathbf{P}_3(\mathbb{R}) : f(0) = f(1), f(2) = 0\}$.

- VS8** $W_1, W_2 \leq V$. Prove $W_1 \cap W_2 \leq V$. Is $W_1 \cup W_2$ a subspace? When?
- VS9** Prove: $S \subseteq \text{span}(S)$; $S \subseteq T \Rightarrow \text{span}(S) \subseteq \text{span}(T)$; $\text{span}(\text{span}(S)) = \text{span}(S)$; $\text{span}(S)$ is a subspace (handle $S = \emptyset$).
- VS11** True, false, or sometimes true (justify): $S \subseteq T$ and (a) $S \text{ LI} \Rightarrow T \text{ LI}$; (b) $S \text{ LD} \Rightarrow T \text{ LD}$; (c) $T \text{ LI} \Rightarrow S \text{ LI}$; (d) $T \text{ LD} \Rightarrow S \text{ LD}$.
- VS12** (a) Define a subspace of $M_2(\mathbb{C})$ of dimension 2. (b) Give $S \subseteq \mathbf{P}_3(\mathbb{R})$ with $\text{span} S = \mathbf{P}_3(\mathbb{R})$ and $S \text{ LD}$. (c) $S = \{x_1 \dots x_n\}$ spans V and is LD: what can be said about $\dim V$?
- VS13** True or false with proof/counterexample: (a) $W \leq V$, V infinite-dim. $\Rightarrow W$ infinite-dim.; (b) $\Rightarrow W$ finite-dim.; (c) V finite-dim. $\Rightarrow W$ infinite-dim.; (d) $\Rightarrow W$ finite-dim. (e) Prove $\mathbf{P}(\mathbb{R}) = \bigcup_n \mathbf{P}_n(\mathbb{R})$ is not finite-dimensional.
- VS14** (a) $W = \{A \in M_2(\mathbb{F}) : A^T = A\}$: basis; if $W \cong \mathbb{F}^n$ find n . (b) Explain why $\dim \mathbb{F}^n$, $M_{n \times k}(\mathbb{F})$, $\mathbf{P}_n(\mathbb{F})$ do not depend on $\mathbb{F}$.

6. Linear Independence

- LI1** LI or LD (use an iff setup): (a) $\{(1, i, 0), (2, 1, i), (1, 0, 1)\} \subseteq \mathbb{C}^3$; (b) $\left\{\begin{bmatrix} 1 & i \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} i & 1 \\ 0 & i \end{bmatrix}, \begin{bmatrix} 1+i & 2 \\ 0 & 1+i \end{bmatrix}\right\} \subseteq M_2(\mathbb{C})$; (c) $\{1+x, 2-x+x^2, 1+3x+x^2\} \subseteq \mathbf{P}_2(\mathbb{R})$; (d) $\left\{\begin{bmatrix} 1 & i \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 2i \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\right\}$; (e) $\{I, 2I, 3I\} \subseteq M_2(\mathbb{R})$ (set-up only).
- LI2** Is x in the span? (a) $(3, 4, 1) \in \text{span}\{(1, 1, 0), (2, 1, 1), (1, 2, 0)\}$; (b) $(1+i, 2) \in \text{span}\{(1, i), (2, 1)\} \subseteq \mathbb{C}^2$; (c) $\begin{bmatrix} 1+i & 2 \\ 0 & 1+i \end{bmatrix} \in \text{span}\left\{\begin{bmatrix} 1 & i \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} i & 1 \\ 0 & i \end{bmatrix}\right\}$; (d) $1+2x+x^2 \in \text{span}\{1+x, 1-x+x^2\}$; (e) $1+x+x^2 \in \text{span}\{1+x, x^2, x^3\}$; (f) $2x+1 \in \text{span}\{x^2+3x+2, x+1, x^2+2\}$ (set-up); (g) $x^2-x \in \text{span}\{x^2+1, x^2+x+1\}$ (system, RREF, conclude).
- LI3** V complex; $u, v, w \in V$. Prove $\{u, v, w\} \text{ LI} \iff \{u+v, u+w, v+w\} \text{ LI}$. (Where does $2 \neq 0$ matter?)
- LI4** $S \text{ LI}$, $S' \subsetneq S$, $W = \text{span}(S)$. Prove $\text{span}(S') \neq W$.
- LI5** $S = \{(1, 2, 3), (3, 2, 1), (1, 1, 1)\}$. (a) Write the system for LI and justify it. (b) Gaussian elimination to REF. (c) Decide LI/LD and, if LD, a LI $S' \subseteq S$ that is a basis of $\text{span} S$.
- LI6** $T \in \mathcal{L}(V, W)$. (a) $\{Tv_1 \dots Tv_n\} \text{ LI} \Rightarrow \{v_1 \dots v_n\} \text{ LI}$. (b) T injective and $\{v_i\} \text{ LI} \Rightarrow \{Tv_i\} \text{ LI}$. (c) T preserves linear dependence.

7. Basis and Dimension

- BD1** Prove $\{e_i\}$ is a basis of $\mathbb{F}^n$, $\{E_{ij}\}$ of $M_{n \times k}(\mathbb{F})$, $\{1, x, \dots, x^n\}$ of $\mathbf{P}_n(\mathbb{F})$; state the dimensions.
- BD2** Basis and dimension: (a) $\{A \in M_2(\mathbb{R}) : \text{tr} A = 0\}$; (b) $\{(x, y, z) : x = 2y\}$; (c) $\{f \in \mathbf{P}_3(\mathbb{R}) : f(0) = f(1) = 0\}$; (d) $\{A \in M_2(\mathbb{C}) : A_{21} = 0\}$; (e) $\{A \in M_2(\mathbb{R}) : a+d = 0, b = c\}$; (f) $\{p \in \mathbf{P}_3(\mathbb{R}) : p(0) = p(1), p(2) = 0\}$; (g) $\{(x, y, z, w) : x+y-z = 0, y+w = 0\}$; (h) $\{A \in M_2(\mathbb{R}) : \text{tr} A = 0, A = A^T\}$; (i) $\{A \in M_3(\mathbb{C}) : A = A^*\}$ as a real vector space (why is it not a complex subspace?).
- BD3** Counting arguments: $\{(1, 0, 0), (0, 1, 0), (0, 0, 1), (1, 1, 1)\}$ not LI; $\{(1, 0, 0), (0, 1, 0)\}$ not spanning $\mathbb{R}^3$; five matrices in $M_2(\mathbb{R})$ are LD; five matrices in $M_{3 \times 2}(\mathbb{R})$ do not span; $\{1, x, x^2, x^3\} \subseteq \mathbf{P}_2 \text{ LD}$; $\{1, x\}$ does not span $\mathbf{P}_2$; ten matrices in $M_3(\mathbb{R})$ are not a basis; n polynomials in $\mathbf{P}_n(\mathbb{R})$ are not a basis.
- BD4** Extend the LI set $\left\{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}\right\}$ to a basis of $M_2(\mathbb{R})$ using standard basis vectors; justify the choice (not every E_{ij} works).
- BD5** $\dim V = n$, $S \text{ LI}$ with $|S| = n$. Using the Extension Lemma and Dimension Bound Theorem, prove S is a basis.
- BD6** Find an LI subset with the same span: (a) $\{(1, 2, 1), (2, 4, 2), (1, 1, 0)\}$; (b) $\left\{\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 4 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}\right\}$; (c) $\{1+x, 2+2x, x\}$.

8. Linear Transformations

- LT1** Linear? Prove or disprove: $T(x, y) = (2x-y, x+3y)$; $T(A) = A + A^T$; $T(x, y, z) = x - 2y + z$; $T(x) = x^2$; $T(p) = p(1)$; $T(p) = p'$; $T(x, y) = (0, x+y)$; $T(A) = \text{tr} A + 1$ ($M_2 \rightarrow \mathbb{R}$); $T(A) = A + I$ on $M_2(\mathbb{R})$.
- LT2** Prove for linear T : $T(0) = 0$; $T(-v) = -T(v)$; $T(\sum c_i v_i) = \sum c_i T(v_i)$; LD preserved; $T(U)$ is a subspace for $U \leq V$; $\text{image}(T) \leq W$.
- LT3** (i) $[T]_\alpha^\beta$, (ii) $[v]_\alpha$, (iii) $[T(v)]_\beta$, (iv) verify $[T(v)]_\beta = [T]_\alpha^\beta [v]_\alpha$: (a) $T(x, y) = (2x+y, x-y)$, α standard, $\beta = \{(1, 1), (1, -1)\}$, $v = (3, 1)$; (b) $p \mapsto p'$ on $\mathbf{P}_2$, $\alpha = \{1, x, x^2\}$, $\beta = \{1, x, 1+x^2\}$, $p = 2+3x+x^2$; (c) $T(A) = A + A^T$ on M_2 with $\beta = \{E_{11}, E_{12}+E_{21}, E_{22}, E_{12}-E_{21}\}$, $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$; (d) $T(a+bx+cx^2) = \begin{bmatrix} a & b \\ c & a \end{bmatrix}$ and $T(a, b, c) = \begin{bmatrix} a+b & a-c \\ b & a+b \end{bmatrix}$ in standard bases.
- LT4** $M = \begin{pmatrix} 1 & 2 & 0 \\ 0 & -1 & 3 \\ 2 & 0 & 1 \end{pmatrix}$, α standard of $\mathbb{R}^3$, $\beta = \{1, x, x^2\}$. Define $T_M(a, b, c)$ explicitly.
- LT5** $T : \mathbb{R}^3 \rightarrow \mathbf{P}_2$ linear with $T(1, 2, 3) = 2 + 3x$, $T(1, 1, 1) = x^2$. Find $T(0, 1, 2)$.
- LT6** (a) I and Z are linear on V ; $I \circ T = T \circ I = T$; $T \circ Z = Z$. (b) $(Tp)(x) = \int_0^x p(t) dt$, $\mathbf{P}_n \rightarrow \mathbf{P}_{n+1}$: linear; $T(x^k)$; $(Tp)(0) = 0$.
- LT8** (a) Prove for $A = [T]_\alpha^\beta$: $(b_1, \dots, b_k)$ solves $Ax = 0 \iff b \in \text{null}(A) \iff \sum b_i v_i \in \ker T$; and $w \in \text{image}(T) \iff [w]_\beta \in \text{col}(A)$. (b) Bases of $\ker T$, $\text{image} T$ and rank-nullity for $T(x, y) = (2x+y, x-y)$;

$p \mapsto p'$; $A \mapsto A + A^T$. (c) $T : \mathbf{P}_2(\mathbb{C}) \rightarrow \mathbb{C}^3$, $a+bx+cx^2 \mapsto (a+ib+2c, (1+2i)a+b+(1+i)c, 2ia+(2-i)b+c)$: matrix, null-space basis, $\ker T$, col-space basis, $\text{image} T$, rank-nullity. (d) $A = \begin{bmatrix} 1 & -2 & 0 \\ 3 & 1 & 4 \end{bmatrix}$, $\alpha = \text{std of } \mathbb{R}^2$, $\beta = \{1, x, x^2\}$: define T , find $\ker$, image .

- LT9** $T \in \mathcal{L}(V)$. Can $\text{image} T = \ker T$ for $V = \mathbb{R}^2$? $\mathbb{R}^3$? State the condition on $\dim V$; prove existence when it holds and impossibility otherwise.
- LT10** (a) $T \in \mathcal{L}(V, W)$, V finite-dim.: $\ker T = \ker T^2 \iff \text{image} T = \text{image} T^2$. (b) $V' \leq V$, $W' \leq W$: $T(V')$ and $T^{-1}(W')$ are subspaces. (c) Choose W' with $T^{-1}(W') = \ker T$ and V' with $T(V') = \text{image} T$; state the most general such conditions.
- LT11** Construct: (a) linear $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ with $\ker T = \{0\}$; (b) linear $T : \mathbf{P}_2(\mathbb{R}) \rightarrow \mathbb{R}^3$ that is not an isomorphism; (c) state $\text{image}(P_{(1,2,1)})$.
- LT12** Yes/No with a one-line dimension argument: $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ surjective? $\mathbb{R}^3 \rightarrow \mathbb{R}^2$ injective? $\mathbf{P}_1 \rightarrow \mathbb{R}^2$ surjective? $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ injective? $M_{3 \times 2}(\mathbb{R}) \rightarrow \mathbf{P}_6(\mathbb{R})$ surjective? $V \rightarrow M_{2 \times 4}(\mathbb{R})$ surjective: what for $\dim V$ (and if V infinite)? Rank: $\mathbb{R}^5 \rightarrow \mathbb{R}^3$ with nullity 4; $M_2 \rightarrow \mathbb{R}^4$ with nullity 3; $\mathbf{P}_2 \rightarrow \mathbb{R}^3$ with nullity 0. Compute $\dim \mathcal{L}(\mathbb{R}^5, \mathbb{R}^3)$, $\dim \mathcal{L}(\mathbf{P}_2, \mathbb{R}^3)$, $\dim \mathcal{L}(M_2(\mathbb{R}), \mathbb{R}^4)$, $\dim \mathcal{L}(M_{2 \times 4}(\mathbb{C}), \mathbf{P}_2(\mathbb{C}))$; $\Phi \in \mathcal{L}(\mathcal{L}(\mathbb{R}^2, \mathbb{R}^3), \mathbb{R}^7)$: can Φ be injective? surjective?
- LT13** Prove for $T \in \mathcal{L}(V, W)$: injective $\Rightarrow$ LI preserved; surjective $\Rightarrow$ spanning sets preserved; $(\dim V < \infty)$ injective $\iff \dim \text{image} T = \dim V$; $(\dim W < \infty)$ surjective $\iff \dim \text{image} T = \dim W$. State the equivalent conditions for injectivity. On $\mathbb{F}^{\mathbb{N}}$: right shift is not surjective; left shift is not injective.
- LT14** (a) $S, T \in \mathcal{L}(V, W)$: $S+T, cT \in \mathcal{L}(V, W)$. (b) $R, S, T \in \mathcal{L}(V)$: $T(R+S) = TR + TS$. (c) $R \in \mathcal{L}(U, V)$, $S \in \mathcal{L}(V, W)$, $T \in \mathcal{L}(W, X)$: SR linear and $T(SR) = (TS)R$. (d) $\mathcal{L}(V, W)$ is a vector space.
- LT15** $T : U \rightarrow V$, $S : V \rightarrow W$. (a) ST injective $\Rightarrow T$ injective; ST surjective $\Rightarrow S$ surjective. (b) Give U, V, W, T, S with ST an isomorphism but S not injective and T not surjective. (c) ST injective: S injective $\iff \dim U = \dim V = \dim W$. (d) ST iso $\Rightarrow \dim U \geq \dim V \geq \dim W$.

9. Eigenvalues and Eigenvectors

- EV3** Eigenvalues and eigenspace bases: (a) $T(x, y) = (3x, 2y)$; (b) $T(x, y) = (x+y, y)$ (is it diagonalizable?); (c) $p \mapsto p'$ on $\mathbf{P}_2(\mathbb{R})$ (matrix in $\{1, x, x^2\}$); (d) $A \mapsto A^T$ on $M_2(\mathbb{R})$ (describe eigenspaces).
- EV4** (a) Eigenvalues of $\begin{pmatrix} 2+i & 3 & 0 \\ 0 & -1+2i & 4 \\ 0 & 0 & 5-i \end{pmatrix}$ and an eigenvector for $2+i$. (b) Eigenvalues and eigenspace bases of $A = \begin{pmatrix} 1+i & 1 & 0 \\ 0 & 1 & i \\ 0 & 0 & 2 \end{pmatrix}$. (c) Define eigenvalue of a linear transformation.
- EV5** (a) Define $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with exactly one eigenvalue and give its matrix. (b) T invertible: what is E_0 ? (c) Eigenvalues of $T(x, y, z) = (x+y+z, 2y-3z, 4z)$. (d) For $T(x, y, z) = (2x, y, -2z)$: diagonalizable? invertible? Why?
- EV6** (a) Prove $\overline{zw} = \bar{z}\bar{w}$. (b) Prove $\overline{Av} = \bar{A}\bar{v}$. (c) $A \in M_n(\mathbb{R})$, $Av = \lambda v$, $v \neq 0$: show $A\bar{v} = \bar{\lambda}\bar{v}$ (and $\bar{v} \neq 0$). (d) $\lambda_1 \neq \lambda_2$ eigenvalues of A : prove $E_{\lambda_1} \cap E_{\lambda_2} = \{0\}$.
- EV8** $A = \begin{pmatrix} 1 & -1 & 2 \\ 1 & 0 & -5 \\ 0 & 1 & 3 \end{pmatrix}$: compute $p(\lambda) = \det(A - \lambda I)$ and all eigenvalues (real and complex).
- EV9** P = orthogonal projection of $\mathbb{R}^2$ onto $\text{span}\{(1, 2)\}$. Find $P(e_1), P(e_2)$, $[P]$, eigenvalues, eigenspace bases, and show the eigenspaces are orthogonal. In general, for $u = (a, b) \neq 0$ show u has eigenvalue 1 and $(-b, a)$ eigenvalue 0.

10. Diagonalization (including change of basis)

- EV1** (a) $A = MBM^{-1}$, $p(x) = \sum a_i x^i$: prove $A^r = MB^r M^{-1}$ and $p(A) = Mp(B)M^{-1}$; $p(\text{diag}(d_i)) = \text{diag}(p(d_i))$; $[T]_{\alpha}^{\alpha} = MDM^{-1} \Rightarrow p([T]_{\alpha}^{\alpha}) = Mp(D)M^{-1}$. (b) $p(D)$: $D = \text{diag}(2, 3)$, x^2-4x+1 ; $D = \text{diag}(1, -2, 4)$, x^3-2x+5 ; $D = \text{diag}(0, 1, -1)$, $2x^3+x^2-1$. (c) $p(A)$ with $A = MDM^{-1}$: $M = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $D = \text{diag}(2, 5)$, x^2+1 ; $M = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $D = \text{diag}(-1, 3)$, x^2-2x ; $M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, $D = \text{diag}(1, 2, 4)$, x^2-3x+2 ; $P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $D = \text{diag}(2, 3)$, x^2+x . (d) $T(x, y, z) = (2x, y, -2z)$, $f(x) = x^3-2x+1$: compute $f(T)$.
- EV2** Eigenvalues and a basis of each eigenspace; is it diagonalizable, and if so find P, D with $A = PDP^{-1}$: $\begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$, $\begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$, $\text{diag}(1, 4, -2)$, $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $\begin{pmatrix} 0 & 1 \\ -2 & 3 \end{pmatrix}$, $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$, $\begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

- EV7** $T, S : \mathbb{C}^2 \rightarrow \mathbb{C}^2$, $[T] = \begin{pmatrix} 2 & i \\ -i & 2 \end{pmatrix}$, $[S] = \begin{pmatrix} \frac{5}{2} & \frac{i}{2} \\ \frac{i}{2} & \frac{3}{2} \end{pmatrix}$. (a) Prove T diagonalizable, S not. (b) Eigenbasis β for T .
 (c) $[T]_\beta^\beta$ and $[I]_\beta^\alpha$. (d) $[I]_\alpha^\beta$ by Gauss–Jordan. (e) $[p(T)]$ for $p = x^2 - 3x + 1$. (f) $p(T)$ explicitly.
- CB1** α, β, γ bases of V . Prove: $[v]_\beta = [I]_\alpha^\beta[v]_\alpha$; $[I]_\beta^\alpha = ([I]_\alpha^\beta)^{-1}$; $[I]_\alpha^\gamma = [I]_\beta^\gamma[I]_\alpha^\beta$; $[I]_\beta^\alpha[I]_\alpha^\beta = I$ (via $II = I$); $[T]_\beta^\beta = [I]_\alpha^\beta[T]_\alpha^\alpha[I]_\beta^\alpha$, hence different matrices of one T are similar.
- CB2** (a) $\alpha = \{(1, 0), (0, 1)\}$, $\beta = \{(1, 1), (1, -1)\}$: $[I]_\beta^\alpha, [I]_\alpha^\beta, [v]_\beta$ for $v = (4, 2)$. (b) α standard of $\mathbb{C}^3$, $\beta = \{(1, 0, i), (0, 1, 1), (1, i, 0)\}$: $[I]$'s and $[v]_\beta$, $v = (2, 3, 1)$. (c) $T(x, y) = (x + 2iy, 3x + y)$ on $\mathbb{C}^2$, $\beta = \{(1, i), (1, -i)\}$: $[T]_\alpha^\alpha, [I]$'s, $[T]_\beta^\beta$ by change of basis. (d) $T(p) = p' + p$ on $\mathbf{P}_2$, $\alpha = \{1, x, x^2\}$, $\beta = \{1, 1+x, (1+x)^2\}$: $[I]$'s, $[T]_\alpha^\alpha, [T]_\beta^\beta$.

11. Orthogonality

- OR1** (*axioms*) Prove $\langle x|y \rangle = \sum x_i y_i$ on $\mathbb{R}^n$ is additive and homogeneous in the first slot, symmetric, positive-definite; hence linear in each slot.
- OR2** $W \leq \mathbb{R}^n$, $W^\perp = \{v : \langle v|w \rangle = 0 \ \forall w \in W\}$. Prove $W^\perp \leq \mathbb{R}^n$. (*challenge*) Prove $W \cap W^\perp = \{0\}$ and $W + W^\perp = \mathbb{R}^n$, i.e. $W \oplus W^\perp = \mathbb{R}^n$.
- OR3** $P_v(x) = \frac{\langle v|x \rangle}{\langle v|v \rangle} v$ on $\mathbb{R}^n$. Prove: P_v linear; image = $\text{span}(v)$, $\ker = \text{span}(v)^\perp$; $P_v \circ P_v = P_v$. Reflection $R_v = 2P_v - I$: linear, fixes $\text{span}(v)$, negates $\text{span}(v)^\perp$, $R_v \circ R_v = I$. Rotation A_θ preserves $\langle \cdot | \cdot \rangle$.
- OR4** (*fluency*) $P_{(1,2,1)}(1, -3, 5)$; $P_{(2,-1,2)}(3, 0, -1)$; $P_{(1,1,0)}(4, -2, 3)$; $P_{(0,1,1)}(2, 5, -1)$; rotation $R_{\pi/2}(3, -1)$; $R_\pi(2, 5)$; $R_{\pi/3}(4, 0)$; $R_{\pi/4}(2, -2)$; $R_{3\pi/2}(-1, 3)$; $R_{5\pi/6}(6, 0)$.

12. Least Squares

No new items. None of the 25 sources assesses least squares, Gram–Schmidt, QR, LU, SVD or orthogonal diagonalization; adding material would violate the efficiency criterion.

13. Other (recall and workflow)

- OT1** *Part-A recall list (one mark each)*. Define: field; vector space; $\mathbb{C}$ operations; $\mathcal{F}(\mathbb{R})$ operations; matrix operations; subspace; closure characterization; linear combination; span; sum of sets; LD; LI; redundant vector; linear system; REF; RREF; pivot; free/basic variable; homogeneous/non-homogeneous/inconsistent/independent-consistent/dependent-consistent system; Extension Lemma; basis; Dimension Bound Theorem; dimension; finite/infinite-dimensional; linear transformation; coordinate vector/matrix; matrix-defined map; fundamental theorem of matrices; kernel; nullspace; image; range; rank; nullity; Rank–Nullity; injective; surjective; bijective; inverse; isomorphism; invertible matrix; multilinear; alternating; determinant; Cramer's rule; diagonal; similar; diagonalizable; eigenvalue/eigenvector/eigenspace (matrix and map); change-of-basis matrix.

HIGH-YIELD ADDITIONS

Newly discovered recurring question types (frequency across the 25 files)

- Subspace test on a constraint set**, with zero-vector disproof ($f(0) = 1$, $p(1) = 1$, $\det A = 0$): 9 files. **Basis + dimension of a constraint subspace**: 6 files.
- Set-up-only system with iff justification** (LI / span / spanning set): 6 files.
- Abstract map $\rightarrow$ matrix $\rightarrow$ kernel/image $\rightarrow$ rank-nullity** ($\mathbf{P}_2 \rightarrow M_2$, $\mathbf{P}_2 \rightarrow \mathbb{C}^3$, $M_2 \rightarrow M_2$): 6 files.
- Dimension yes/no** (“can T be surjective/injective?”): 6 files. **Matrix inverse $\rightarrow$ explicit T^{-1}** : 4 files.
- Part-A “construct/define an example”**: ≥ 11 items across quizzes and mocks.
- Determinant from axioms**; $p(A)$ via $A = PDP^{-1}$; **conjugate transpose expansion**: each ≥ 3 files.
- Complex entries in every computation** (RREF, inverse, eigenvectors, products): $\approx 40\%$ of computational items.

Repeated assessment patterns

- Structure: quiz = 15 marks (Part A 3×1 , Part B 12); mock midterm 50 (+Part C 3–4); mock final 70 (+Part C 4); all no-calculator.
- The instructor states quizzes are “VERY SIMILAR” to the homework, and they are: 82 of 92 quiz items repeat a homework task. Practise homework wording, not new variants.

- Reused objects**: $T(x, y, z) = (x+y, y+z, x+z)$; $T(A) = A + A^T$; $p \mapsto p'$; $T(x, y) = (2x+y, x-y)$; $\begin{pmatrix} 1+i & 1 & 0 \\ 0 & 1 & i \\ 0 & 0 & 2 \end{pmatrix}$;

the four systems of Assignment 3a (used five times). Master these five maps and two matrices.

- Every Part B proof asks for: *given* / *to prove* / *technique*; quizzes add “using an if-and-only-if argument”.
- Week-themed Part A recall: one definition or theorem statement per quiz.

Common instructor traps

- Column-space basis from **original** pivot columns, not RREF columns; row-space basis from RREF rows.
- $\dim \mathbf{P}_n = n + 1$; $\dim M_{3 \times 2} = 6$; $\dim_{\mathbb{C}}$ vs $\dim_{\mathbb{R}}$ for Hermitian matrices.
- Conjugate the scalar: $(cX)^* = \bar{c}X^*$; $\bar{\bar{i}} = -i$; $(AB)^* = B^*A^*$ reverses order.
- “Surjective $\Rightarrow \dim V \geq \dim W$ ” but V may be infinite-dimensional; injective/surjective are not equivalent off finite dimension.
- Left-identity only: uniqueness needs commutativity (MT1 B2). Zero vector is $f(x) = 0$, never the number 0.
- $\dim(\text{span}) \leq n - 1$ for a dependent spanning list of size n (not n).
- Basis extension: not every E_{ij} keeps independence. Constant term of $p(A)$ is a_0I , not a_0 .
- Field matters: $2 = 0$ in characteristic 2 (Q8); Hermitian set is a real, not complex, subspace.
- Real- A eigenpairs: $(\lambda, v) \Rightarrow (\bar{\lambda}, \bar{v})$; complex roots appear in 3×3 real matrices (Final B7: $2, 1 \pm 2i$).

Grading-sensitive mistakes (marks stated in sources)

- Set-up problems: marks for the **iff justification**, not the solution.
- Determinants “using multilinear/alternating properties”: using $ad - bc$, cofactors or triangular shortcut without justification scores zero (explicit on Q10, Final A9, B5).
- Cramer’s rule: must **compute** $\det A \neq 0$ to justify use; answer in form $a + bi$ for complex arithmetic.
- Basis claims need both **spanning and independence**; standard basis must be written out before quoting its dimension (Q5 B2).
- Definitions must name underlying set, operations and all axioms (Q1 A3); state the field.
- Proofs of existence (identity in $\mathbb{C}$) must show the “for all” check (Q1 B2); disproofs need an explicit counterexample or the zero-vector failure.
- Eigenvectors must be non-zero; show $\bar{v} \neq 0$ (Final B2c).
- “Show your work” on Part A construction items; final-answer-only earns no credit.

Recommended minimal integration (no rewrite of existing lessons)

W1: VS1, VS5, OT1. **W2:** MO1–MO3, MO5, VS3, VS6 + “abstract spaces” intro (D2). **W3:** RR1–RR5, LS1, LS5. **W6:** LS2, LI1–LI6. **W8:** VS7–VS14, BD1–BD6. **W9:** LT3 (coordinate verification). **W10–11:** LT1–LT14. **W12:** INV1–INV3, LT13–LT15, D6. **W13:** DT1–DT7. **W14:** EV1–EV9, CB1–CB2. **W15:** add MO6, LI5, VS2, VS12, LT4–LT5, LT11, EV5, EV7–EV8 as mock material. Total addition: 83 entries, of which 52 Core.